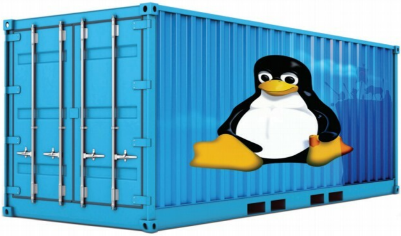
The Usefulness of Linux Containers (LXC)

Containers are the future when it comes to using and shipping applications. But Linux containers or LXC can be used for more than just that. This article covers the use of LXC on a daily basis as well as for production.



Although containers have become popular due to their extensive use of Docker by Docker Inc., which has been in the market since 2013, Linux has also had its own containers for several years now. These native Linux containers are also referred to as LXC. The LXC project has been around since 2008, and is being actively maintained and supported by Canonical Ltd. It is a set of tools (technically, more an API) which gives you an interface to create a container using Linux namespaces and cgroups (Control Groups). Linux namespaces (mnt, pid, net, ipc, uts, user, etc) is a feature of the kernel that is used for a different type of isolation, whereas cgroups is used to set limitations over the resources (mostly hardware like memory and disk size) used by any process.

Mount namespace (mnt): In this namespace, a process views different mount points other than the original system mount point. It creates a separate file system tree associated with different processes, which restricts them from making changes to the root file system.

PID namespace (pid): PID namespace isolates a process ID from the main PID hierarchy. A process inside a PID namespace can have the same PID as a process outside it, and even inside the namespace, you can have a different init with PID 1.

UTS namespace: In the UTS (UNIX Time-sharing System) namespace, a process can have a different set of domain names and host names than the main system. It uses *sethostname()* and *setdomainname()* to do that.

IPC namespace: This is used for inter-process communication resources isolation and POSIX message queues.

User namespace (user): This isolates user and group IDs inside a namespace, which is allowed to have the same UID or GID in the namespace as in the host machine. In your system, unprivileged processes can create user namespaces in which they have full privileges.

Network namespace (net): Inside this namespace, processes can have different network stacks, i.e., different network devices, IP addresses, routing tables, etc.

LXC is a set of tools that makes it possible to use these features and create something called a container, which is nothing but a very lightweight VM with less isolation. The reason for less isolation is the absence of a hypervisor layer, which makes containers use the same kernel as the host system. But this absence also makes containers more widely usable, as there is no overhead of configuring and no need for a hypervisor; so you can select which resources you wish to isolate according to your usage. Let's now dive into the Linux container world.

Getting started with LXC

Note: For demonstration purposes, I will use LXC in Ubuntu 16.04. LXC has great support on Ubuntu since the LXC project is backed by Canonical Ltd, which is also the publisher of the Ubuntu operating system. But you can also install LXC on any other Linux distribution through its official repositories.

In your Ubuntu machine, install LXC and *lxc-templates* by pasting the following command in your terminal:

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$sudo apt install lxc lxc-templates
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